

MIKHIL GEORGE EDATTEL

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EDUCATION

University of Notre Dame

B.S. in Mechanical Engineering

Concentration: Thermal & Fluid Sciences, Computational Engineering

August 2024 - May 2028

GPA: 3.73/4.00

University of Notre Dame London – Summer Engineering Study Abroad

May 2025 – June 2025

Honors: Certified SolidWorks Associate (CSWA) · President's Circle Mentorship Scholar · Dean's List

EXPERIENCE

Cooper Standard – CAE

CAE Engineer Intern

Northville, MI

June 2026 – August 2026

- Achieved a 10 N compression load deflection (CLD) target for a Ford F-150 Lightning outer-belt seal by iterating geometry in MSC Marc/Mentat using nonlinear 2D FEA across ± 2 mm door variation and +2.5 mm overslam conditions
- Determined optimal flange thickness for a Toyota truck sealing application by comparing peak insertion and extraction forces across 0.95, 1.50, and 2.75 mm thicknesses using FEA and documenting recommendations in a technical report
- Evaluated GM Silverado glass-run seal stability using 3D nonlinear contact FEA in MSC Marc/Mentat, modeling Santoprene and Celloprene installation and predicting buckling, rippling, and bending under overcompression

Domer Rover – Mars Rover Design Team

Project Manager

Notre Dame, IN

September 2026 – Present

- Direct development of a University Rover Challenge rover for a 30+ member team, coordinating mechanical, electrical, autonomy, and communications work around mission requirements and subsystem dependencies
- Lead design reviews and evaluate technical trade-offs for arm and astrobiology modules, resolving conflicting subsystem needs and selecting configurations compatible with the integrated rover
- Redesigned central housing unit from acrylic to aluminum to integrate electronics, computing, and power systems while enabling interchangeable arm and science modules; verifying support of a 70 lb static load through physical testing
- Conducted center-of-mass analysis to guide housing placement relative to the suspension and tested rover mobility across sand, gravel, and rock to identify needed mechanical improvements

RESEARCH

Hypersonic Systems Lab – University of Notre Dame

Research Assistant

Notre Dame, IN

August 2026 – Present

- Developing an in-house CFD solver by implementing finite volume methods for hyperbolic PDEs in MATLAB and C++, targeting replacement of a commercial ANSYS Fluent workflow for research purposes
- Studying finite volume methods for hyperbolic conservation laws using Randall J. LeVeque's text

H.B. Chang Lab – University of Notre Dame

Research Assistant

Notre Dame, IN

August 2025 – May 2026

- Miniaturized benchtop rotary nanofabrication system into 500 g handheld device by redesigning solution delivery, airflow mechanisms, and power systems.
- Resolved rotor instability at 10,000 RPM by redesigning the turbine-spinneret coupling in SolidWorks, correcting alignment, and validating performance across 20+ experimental runs
- Designed an annular air-jet ring to guide spun fibers toward a collection plate, achieving 95% collection efficiency in controlled deposition trials

PROJECT

Servo-Controlled Marble Maze

March 2026 – May 2026

- Led mechanical design and build planning for a two-axis marble maze, selecting a nested gimbal architecture to achieve $\pm 20^\circ$ tilt on both axes
- Developed SolidWorks models and dimensioned manufacturing drawings, then fabricated and assembled a laser-cut plywood frame, a 3D-printed maze, and dual-servo actuation hardware into a functioning prototype

SKILLS

Simulation & Analysis:

MSC Marc/Mentat · MATLAB · Ansys Fluent · Nonlinear Contact Analysis · Hyperelastic Material Modeling

CAD & Design:

SolidWorks (CSWA) · Siemens NX · Engineering Drawings · Dimensional Tolerancing

Programming:

C/C++ · Python · Arduino

Manufacturing:

Additive Manufacturing · Laser Cutting · MIG Welding · Waterjet Cutting · Soldering